

Pre-Live Validation Report

Gold Trend — Strategy | XAUUSD 30m | eight checks run against the research engine, 3 August 2026

#	Check	Result	Verdict
1	Buy and hold	+5,922% at 33.80% DD vs gold +179.1% at 29.08% (33x)	PASS
2	Long vs short	shorts PF 1.891 vs longs 1.806; 35.1% of net profit	PASS
3	Per-year / out-of-sample	train PF 1.351, OOS 2024-26 PF 1.976 at 20.80% DD	PASS*
4	Parameter robustness	Donchian flat (+/-3%); trail is a sharp ridge (-19%)	MIXED
5	Fill realism	already hardcoded in the script	PASS
6	Repaint	no security calls, all lookbacks [1]-offset	PASS
7	Sanity: capital / leverage	\$25,000 minimum at current gold prices	ACTION
8	Monte Carlo (added)	actual 33.80% DD is better than the median reshuffle	CAUTION

*The out-of-sample half is the stronger one, which is the right direction. But every parameter in this build was selected knowing the full sample, so 2024–26 is not virgin data. Deflated Sharpe (0.9996) and probability of backtest overfitting (0.099) are the defences against that and both pass. Forward performance from today is the only true out-of-sample this project will ever get.

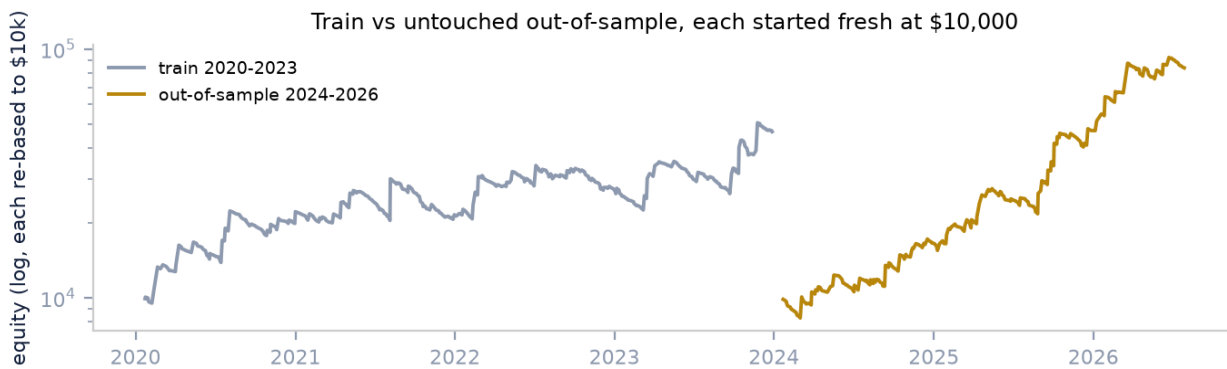
1–2. Does it beat holding gold, and is the short side worth keeping?

	Net return	Max drawdown	Profit factor
Strategy	+5,922%	33.80%	1.834
Buy and hold gold	+179.1%	29.08%	-
Outperformance	+5,743 pp		33.1x

Side	Trades	Win rate	Profit factor	Net P&L	Share of profit
LONG	484	24.4%	1.806	+\$384,477	64.9%
SHORT	253	16.6%	1.891	+\$207,712	35.1%

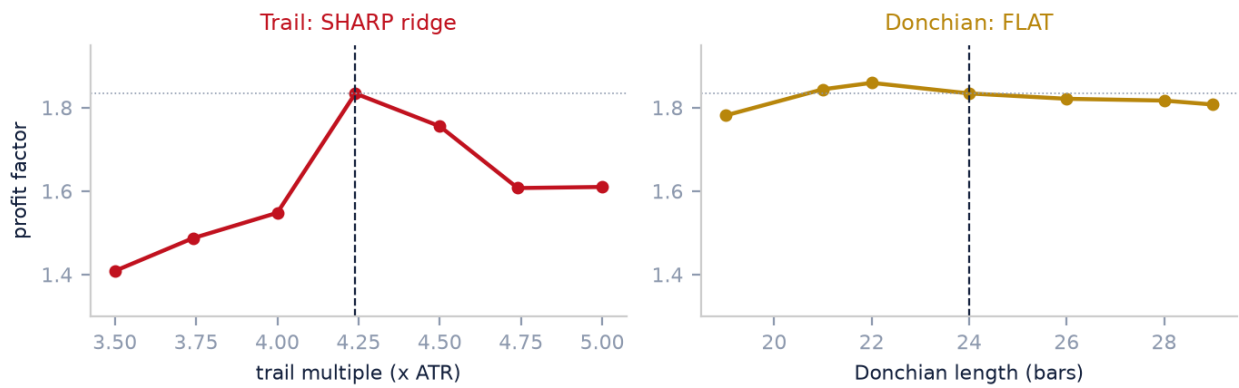
The short side is the BETTER book, not dead weight. It has a higher profit factor than the long side (1.891 against 1.806) despite a lower win rate, and supplies just over a third of net profit. Cutting or shrinking it was tested separately and rejected: it degrades performance in exactly the two years gold fell, and it is harmful on US30.

3. Out-of-sample split



Window	Trades	Profit factor	Net return	Max drawdown	CAGR
TRAIN 2020-01 to 2023-12	440	1.351	+365.2%	33.79%	47.0%
OOS 2024-01 to 2026-07	285	1.976	+739.4%	20.80%	128.6%
Full sample	737	1.834	+5,921.9%	33.80%	85.1%

4. Parameter robustness — and it is not symmetric



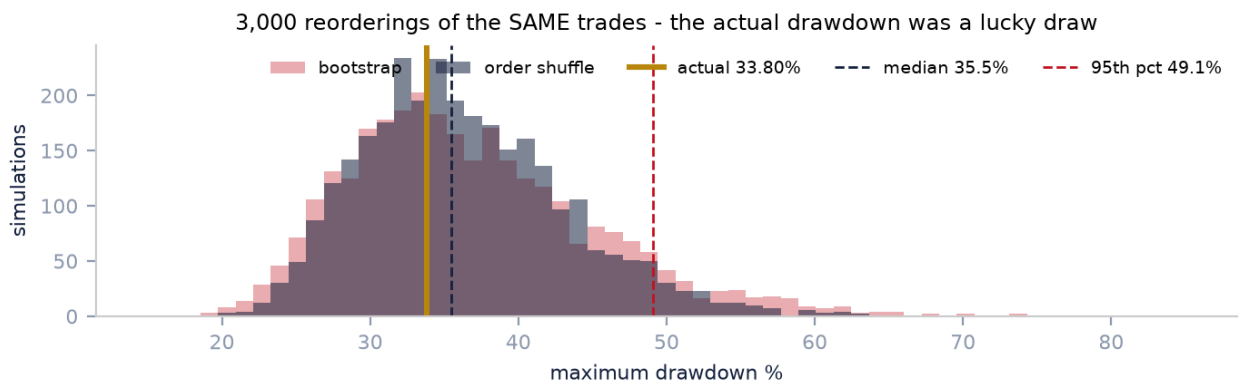
Parameter	Range tested	Profit factor range	Spread	Verdict
Donchian length	19 - 29 (24 +/- 20%)	1.781 - 1.859	+/- 3%	flat, safe
Trail multiple	3.74 - 4.74 (4.24 +/- 0.5)	1.418 - 1.859	-19% / -12%	sharp ridge

Trail multiple	3.50	3.74	4.00	4.24	4.50	4.74	5.00
PF, coupled	1.409	1.487	1.548	1.834	1.755	1.607	1.610

Pinning position size while varying only the trail gives the same shape (1.411 / 1.498 / 1.549 / **1.834** / 1.768 / 1.631 / 1.618), so this is a genuine ridge and not an artifact of the trail also setting position size. **All 15 tested combinations remain profitable and none inverts** — the strategy degrades off the peak rather than breaking.

Practical consequence. The entry length can be wrong by 20% and you would barely notice. The trail cannot. If your broker's ATR differs from OANDA's — different feed, different session boundaries, different spread handling — the effective trail width shifts and this is the parameter that will feel it. Verify ATR(14) on your own feed against 13.2 points at gold \$4,081 before trusting the shipped 4.24.

8. Monte Carlo — the most useful check on the list



Per-trade returns were resampled as a fraction of equity at the time of each trade. Shuffling raw dollar P&L would be wrong under compounding — late trades are roughly forty times larger than early ones, so a dollar shuffle would simply move the big trades around and flatter the result.

Measure	Actual	Median	95th percentile	Worst of 3,000
Max drawdown - order shuffled	33.80%	35.5%	49.1%	71.2%
Max drawdown - bootstrapped	33.80%	35.5%	52.5%	83.1%

Outcome	Frequency
Reorderings exceeding a 50% drawdown	3.8%
Reorderings exceeding a 60% drawdown	0.3%
Bootstrap runs ending negative	1 in 3,000 (0.03%)
Bootstrap net return, 5th percentile	+721%
Bootstrap net return, median	+5,913%

Your 33.80% drawdown was a fortunate ordering, not a ceiling. The median reshuffle of the very same trades produces 35.5%, the 95th percentile 49.1%, and the worst 71.2%. **Size the account for a 50% drawdown, not for 33.8%.** On \$25,000 that is a \$12,500 loss to sit through, not \$8,450. Nothing would be broken if it happened — it is inside the normal range of this trade distribution.

5–6. Fill realism and repaint

Both were audited in the source rather than by eye. The settings the checklist asks you to toggle are already fixed in the script.

Setting	Required	In the script
Fill orders on bar close	ON	process_orders_on_close = true (line 177)
Recalculate on every tick	OFF	calc_on_every_tick = false (line 183)
Slippage	broker's real spread	200 ticks = 0.20 points (line 176) - CHANGE THIS
Commission	broker's real cost	0.07 per contract (line 163)
Bar magnifier	your choice	TradingView-side toggle; recommended ON

Repaint audit

Construct	Count	Why it matters
request.security	0	no higher-timeframe calls, so no lookahead vector exists
barstate.isrealtime / varip / timenow	0	no branch behaves differently in real time
ta.highest / ta.lowest	4, all [1]-offset	a bar cannot set the trigger it then crosses
chandelier anchor	high[1] / low[1]	a bar cannot move its own stop and then hit it
bestPx (excursion tracker)	after strategy.exit	the excursion always reads through the previous bar

The only textual match for the word "lookahead" in the file is a comment recording BUG-019, a defect of this exact kind found and fixed earlier in this project. **Nothing in this script can repaint.** If the reload test on TradingView does show changed statistics, the cause is data revision on your broker's feed, not the code.

7. Sanity — capital and leverage

Account	Position (oz)	Peak stack	Notional at peak	Verdict
\$5,000	0.90	4.48 oz	\$18,271	Broken - takes ~9% of signals
\$10,000	1.79	8.95 oz	\$36,542	Works, leaks ~1/3 of return
\$25,000	4.48	22.39 oz	\$91,356	Recommended minimum
\$50,000	8.95	44.77 oz	\$182,712	Full fidelity

Peak stack is the position after all four adds, about 3.65x account leverage. At 1:100 broker leverage that requires 3.7% of the account as margin; at 1:20 it requires 18.3%. Margin is not the binding constraint — whole-ounce granularity is.

What still has to be done manually on TradingView

Step	Why it cannot be answered here
Reload the chart and re-add the strategy; confirm signals and statistics do not change	tests TradingView's own data handling and your broker feed, not the script logic
Turn the bar magnifier on and re-run	changes how TradingView resolves intrabar fill order; the research engine models this differently
Set initial capital and leverage to match your real account	the shipped \$10,000 is a research default, not your position sizing
Re-run at your broker's real gold spread	the single largest risk to this strategy; break-even is 1.2-1.5 points

Generated 3 August 2026 from 78,695 bars of XAUUSD 30-minute data, 737 positions, 1 Dec 2019 to 30 Jul 2026. All figures include 0.07 commission and 0.20 points of slippage per side. Past backtested performance is not a prediction. Research record, not investment advice.